☑ Core Features Your News Aggregator Should Have

Here's a breakdown of **must-have features**, grouped by function:

🧠 A. Search, Filtering & Recommendation

- **Keyword/topic search** (e.g., "LGBT", "Gaza", "AI regulation")
- **Topic suggestion / autocomplete**
- **Filter by**:
  - Date range
  - Language
  - Country/Region
  - Source (e.g., BBC, Dawn)
  - Bias level (if implemented)
- **Related topics**: Show trending or connected topics
- **Personalized recommendations** (based on user interests or history)

📰 B. News Presentation

- ✅ **Aggregated view**: Group multiple articles on the same story (deduplicated)
- ✅ **Auto summary**: Provide a short **AI-generated summary** of multiple articles
  - Example: "7 sources confirm ceasefire talks are stalled. 3 highlight ongoing bombings in Gaza."
- ✅ **One-paragraph digest**: Use LLM (like GPT) to generate one neutral summary per news cluster
- ✅ **Bias-aware titles** (e.g., "5 different headlines" to see bias variation)
- ✅ **Timeline view**: See how the story evolved over time
- ✅ **Source diversity**: See the same story across local/international sources

📊 C. User Features

- User login/signup (email/social)
- **Favorites/bookmarks**
- **History tracking** (searched, read, clicked)
- Feedback or flagging for misleading news
- Profile-based news feed
- **"My Bias Bubble" detector**: optional analysis showing what ideological bubble user is in

🌍 D. Cultural Sensitivity & Coverage Gaps

- **Coverage Map**: Show which regions/sources are covering the topic
- **Topic Gaps**: Flag if a source avoids a sensitive topic (e.g., no LGBT reporting)
- **Sentiment/Bias Score** per article
- Tags like:
    - Underreported
    - Bias Detected
    - Sensitive Content

🔒 Legal Aspect: Can You Summarize News Articles?

Yes, **with some conditions**:

What You Want to Do	Is It Legal?	Explanation
Show title, link, source	✅ Yes	This is metadata — fair use
Show short snippet (1–2 sentences)	✅ Yes	Under "fair use" or "quotation" in most jurisdictions
AI-generated **summary of 1 article**	⚠️ Caution	Safe if brief & links to source. Avoid large quotes verbatim
AI-generated **summary of multiple articles**	✅ Safer	Because you're *transforming* the content, not copying
Show full content	❌ No	Only if you have license or it's open/public domain

☑ Safe Practice Suggestion:

- **Crawl metadata** only (title, date, source, URL)
  Use LLM to summarize multiple sources in **your own words(LLM** stands for **Large Language Model**.It's a type of **artificial intelligence (AI)** model that is trained on massive amounts of text data to understand and generate human-like language.)
- Always **link to original source** for full reading
- Include **source logo or name** to credit

🧠 Bonus Features (Advanced)

- **AI-driven topic clustering**
- **News map**: Show where news is coming from on a world map
- **Emotion detection**: Show whether a news story triggers anger, fear, hope, etc.
- **Fake news detection** (e.g., based on source credibility or NLP signals)

🧪 Example User Flow

A user searches **"Gaza ceasefire"**
The aggregator:

1. Detects cluster of 30+ articles
2. Removes duplicates or near-duplicates
3. Summarizes the main narrative in one paragraph (with GPT/NLP)
4. Shows 5-10 individual source headlines below
5. Provides timeline & sentiment analysis
6. Gives a "coverage bias" heatmap (e.g., Al Jazeera focuses on civilian deaths, Fox News on Hamas)

🧑 USER'S PRESPECTIVE:

As a user, here's what I want in a news aggregator:

🗞️ 1. Personalized News Feed

I don't want to scroll through 100 random articles. I want the feed to **learn from what I read, like, save, or search**.

- ✅ Buildable with: User behavior tracking + tagging system + recommendation engine
- 🔧 Tools: MySQL tables for history, favorites, tags; Python-based recommender system

🔍 2. Smart Search

If I search "Gaza", I want all relevant news — even if articles use words like "Middle East conflict" or "Palestinian war".

- ✅ Buildable with: NLP keyword expansion + synonym mapping + full-text search
- 🔧 Tools: MySQL FULLTEXT, Elasticsearch, or embedding-based vector search

🧠 3. Auto-Summary of News

I don't have time to read 10 articles. Just give me **1 paragraph that summarizes all of them** on that topic.

- ✅ Buildable with: LLM summarization
- 🔧 Tools: GPT API / LLaMA / Open-source summarizer models

📊 4. Viewpoints Comparison

I want to compare how **BBC**, **Al Jazeera**, and **Fox News** cover the same story differently.

- ✅ Buildable with: News clustering + source labeling
- 🔧 Tools: Topic clustering using NLP, tagging system in DB, UI component for comparing articles side-by-side

💖 5. Bookmark, Like, and Follow Topics

I want to **bookmark articles**, mark **favorites**, and **follow tags** like "Women", "Climate", "Tech", "LGBT".

- ✅ Buildable with: Tags, favorites, topic following, user preferences table
- 🔧 Tools: MySQL relationships + UI buttons + session storage

📬 6. Email Notifications (Optional)

Send me a daily or weekly **summary of top stories I follow**.

- ✅ Buildable with: Scheduled tasks + user preferences
- 🔧 Tools: Cron jobs, Python script, SMTP email, or third-party email service

🌐 7. Multi-Language Support

I want to read news in **English**, but also **Urdu** or **Arabic**, depending on region/source.

- ✅ Buildable with: Language detection + translation option
- 🔧 Tools: LLMs, Google Translate API, langdetect libraries

🚫 8. Filter Biased or Sensitive Topics

I want an option to **hide certain topics** like war, or filter **biased sources**.

- ✅ Buildable with: Sentiment analysis + topic filters
- 🔧 Tools: NLP classifiers, user preferences, source credibility ratings

🗺️ 9. Global + Local Mix

I want to see what's happening in **my city**, **my country**, and **globally** — in one place.

- ✅ Buildable with: Location-aware crawler or tagging
- 🔧 Tools: Geo-IP, user location, and source categories (Local/Regional/Global)

🕰️ 10. Trending & Archive Views

I want to see **what's trending today**, or go back and read about "Gaza conflict in Jan 2024".

- ✅ Buildable with: Date-based filtering, topic popularity metrics
- 🔧 Tools: MySQL timestamped data + analytics engine

🧠 BONUS: "Explain Like I'm 12" Button

When reading complex news, give me an **AI-powered simple explanation** of what's happening.

- ✅ Buildable with: LLM simplification
- 🔧 Tools: GPT prompt: "Explain the following news article in simple terms"

☑ Database Tables to Support This

Here's how your DB supports these features:

Table Name	Purpose
news_articles	Stores scraped news content
sources	Stores metadata about each news site
tags	Holds topics like "LGBT", "Tech", "Politics"
article_tags	Many-to-many relation for tagging articles
users	Stores user info (email, prefs)
favorites	Tracks saved/bookmarked articles
user_history	Stores search and click behavior
topic_synonyms	Maps terms like "Tech" = "Science & Tech"
summaries	(Optional) Stores generated summaries
feedback	Stores user ratings on article accuracy/bias

☑ Final Thought

As a user, I want:

- **Control** (filters, bookmarks, personalization)
- **Clarity** (summaries, bias detection)
- **Trust** (source transparency)
- **Discovery** (explore new sources/topics easily)

TOOLS USED:

MySQL + Python + open-source NLP + GPT/LLM integration.